Description of Prototype

This part includes the system requirements, the preliminary design, and how the system is being evaluated and tested. SanctiWalk is delivered as two Android applications and one web application: the pilgrim application, which is a React and TypeScript web application packaged for Android; the augmented reality heritage tour, which is a separate Unity application using Google ARCore; and the parish administration console, which runs in a web browser. The augmented reality tour is a separate application by necessity, because ARCore's world tracking keeps a station anchored to a place after the printed trigger image has left the camera's view, and no web browser interface presently offers that capability.

The requirements below are stated separately for the pilgrim application and for the augmented reality tour, because the tour requires substantially more of the device than the rest of the system does. A visitor whose handset meets only the first set may use every part of SanctiWalk except the augmented reality tour.

• The system will have the following minimum system requirements:

For the pilgrim application, excluding the augmented reality tour:

CPU: Quad-core ARM processor or similar

Camera: Any rear-facing camera, required only for the scanner

RAM: At least 2 GB of RAM

Storage: 150 MB of available storage

Operating System: Android 7.0 (Nougat) or greater

Network: An internet connection for the scanner, the parish assistant, the map and live parish updates

For the augmented reality heritage tour:

CPU: 64-bit ARM processor (arm64-v8a); 32-bit devices are not supported

Camera: ARCore-compatible camera

RAM: At least 4 GB of RAM

Storage: 600 MB of available storage

Operating System: Android 10 or greater

Required service: Google Play Services for AR (ARCore), with the device listed as ARCore-certified by Google

Network: An internet connection, required for visual positioning

ARCore certification is the requirement that cannot be substituted. A handset that satisfies every other specification but does not appear on Google's list of ARCore-certified devices cannot run the augmented reality tour at all.

• The system will have the following recommended system requirements:

CPU: Snapdragon 695 Octa-core Max 2.2GHz or similar

Camera: 12 MP or greater with autofocus and ARCore support

RAM: 6 GB

Storage: 2 GB of available storage

Operating System: Android 13

Network: A stable mobile data or Wi-Fi connection

Android 13 is recommended for a reason beyond performance. From that release the operating system prompts the user for notification permission, upon which the Mass reminders and the subscription alerts for ministries and sacraments depend.

• The system will have the following optimal system requirements:

CPU: Snapdragon 8 Gen 2 or greater

Camera: 50 MP or greater with ARCore depth support

RAM: 8 GB or more

Storage: 4 GB or more of available storage

Operating System: Android 14 or greater

Beyond these specifications the proponents observed no further benefit. The system is not demanding at the upper end, and a more powerful handset principally shortens the initial visual-positioning step at the beginning of the augmented reality tour.

• Basis of the memory requirement:

The augmented reality tour renders photogrammetry models of the parishes' statues and of the old San Roque church. The model of San Roque comprises 1,614,065 vertices, the Santo Nino de Pajotan 1,551,117, the old San Roque church 1,094,026, and Mary Help of Christians 1,055,770, each carrying a 4096 by 4096 pixel colour texture. A model of this size occupies approximately 50 MB of vertex data before its texture is loaded, and one station may hold a model in memory while a neighbouring station's model is still resident. This is the basis of the 4 GB minimum and of the recommendation of more.

• Permissions required from the user:

The application requests each permission at the moment it is needed rather than on first launch. The camera is required by the scanner and by the augmented reality tour. Location is required to detect which parish the visitor is at and to provide walking directions. Microphone access is required for the Rosary to follow the user's voice. Notification permission is required for Mass reminders and for the subscription alerts.

Internet access is required by the scanner, the parish assistant, the map and the augmented reality positioning. Where a permission is declined the affected feature states so and the remainder of the system continues to operate.

• Services required on the device:

In addition to Google Play Services for AR, the system depends on services installed on the handset rather than supplied by the application. Google Play Services is required for Google Sign-In. A text-to-speech engine is required for the narration of church history, of scanner results and of the guided Rosary. A speech-recognition service is required for the Rosary to follow the user as they pray.

Filipino voice data for the text-to-speech engine warrants particular mention. The application requests a Filipino voice correctly, but where the handset has not downloaded Filipino voice data the engine substitutes its default voice, and Tagalog text is then read aloud with English pronunciation. The voice data is installed from the device's own text-to-speech settings and not from within SanctiWalk.

• Conditions required at the parish:

The augmented reality tour additionally depends on conditions in the church itself. The parish's printed A4 trigger sheet must be mounted at the position recorded during the survey, because every station's position is computed relative to that sheet. The sheet must be printed at full scale rather than fitted to the page, because the application is informed that the sheet measures 210 by 297 millimetres and derives distance from that measurement. Adequate lighting is required for ARCore to recognise the image and track the room, and a mobile data or Wi-Fi signal is required inside the building because visual positioning is performed over the network.

• Server-side requirements:

The system requires a Node.js runtime hosted on a service providing HTTPS for the scanner and parish assistant endpoints; a Google Firebase project providing Authentication, Cloud Firestore and Cloud Storage; an application programming interface key for the generative model used by the scanner and the parish assistant; credentials for the visual positioning service used by the augmented reality tour; and a static web host serving the pilgrim application, the download page and the administration console.